

## PYTHON INTERVIEW QUESTIONS

1. What is Python?
2. What are Python data types?
3. Difference between List and Tuple.
4. What is a Dictionary?
5. Explain OOP concepts in Python.
6. What is inheritance?
7. What is polymorphism?
8. What are decorators?
9. What are generators?
10. What is exception handling?

## SQL INTERVIEW QUESTIONS

1. What is SQL?
2. Difference between WHERE and HAVING.
3. What is a JOIN?
4. Explain INNER JOIN.
5. Explain LEFT JOIN.
6. What is a Primary Key?
7. What is a Foreign Key?
8. What is normalization?
9. What is a subquery?
10. What is a view?

## MACHINE LEARNING INTERVIEW QUESTIONS

1. What is Machine Learning?
2. Types of Machine Learning.
3. Difference between Supervised and Unsupervised Learning.
4. What is Overfitting?
5. What is Underfitting?
6. What is Cross Validation?
7. What is Feature Engineering?
8. What is Random Forest?
9. What is XGBoost?
10. What is Gradient Descent?

## NLP INTERVIEW QUESTIONS

1. What is NLP?
2. What is Tokenization?
3. What are Stopwords?
4. Difference between Stemming and Lemmatization.
5. What is TF-IDF?
6. What is Word2Vec?
7. What is POS Tagging?
8. What is Named Entity Recognition?
9. What is Sentiment Analysis?
10. What are Embeddings?

## DEEP LEARNING INTERVIEW QUESTIONS

1. What is Deep Learning?
2. What is a Neural Network?
3. What is an Activation Function?
4. What is ReLU?
5. What is Backpropagation?
6. What is an RNN?
7. What is LSTM?
8. What is GRU?
9. What is CNN?
10. What is Vanishing Gradient Problem?

## TRANSFORMERS AND LLMs

1. What is a Transformer?
2. What is Self Attention?
3. What is Multi Head Attention?
4. What is Positional Encoding?
5. Difference between Encoder and Decoder.
6. What is BERT?
7. What is GPT?
8. Difference between BERT and GPT.
9. What is Fine Tuning?
10. What is Prompt Engineering?

## GENAI, RAG AND LANGCHAIN

1. What is Generative AI?
2. What is Retrieval Augmented Generation (RAG)?
3. Explain RAG Architecture.
4. What are Embeddings?
5. What is a Vector Database?
6. What is ChromaDB?
7. What is LangChain?
8. What are Chains in LangChain?
9. What is Memory in LangChain?
10. Explain a real-world RAG use case.